

Sick-it Learn Recreation and Implementation

CSE299-sec-5

Group-5

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****Week 1 Report - Foundation & Architecture****

During Week 1 of the project, the primary focus was on establishing a robust foundation and project architecture. Key accomplishments included setting up a modular project structure and building a functional command-line interface (CLI) in `main.py`, enabling users to specify datasets, models, and options for execution. Configuration management and a logging system were implemented to support scalability and maintainability. Core infrastructure components such as data loading utilities (`utils/data_loader.py`), a result-saving mechanism (`utils/results.py`), and base model interfaces (`models/base_model.py`) were developed. Additionally, a Linear Regression model was built from scratch (`models/linear_regression.py`), marking the completion of the project's first core algorithm. Auxiliary scripts like `setup_week1.sh` and `demo_week1.sh` were also created to facilitate easy setup and demonstration workflows. All deliverables were tested and verified, including outputs stored in structured directories like `RESULTS/`. The week concluded with a fully functional CLI and foundational systems ready for the upcoming work on regression models and evaluation metrics.